

End Behavior and Multiplicity: Sketching Polynomials

Answer Key

Precalculus — Polynomial Function Analysis

Quick Answers

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|-----------------------|--|
| 1. $\infty, -\infty$ | 6. $-\infty, \infty$ |
| 2. $-\infty, -\infty$ | 7. 8, -8 |
| 3. -3, 1 | 8. $(x-2)(x-1)(x+1)(x+2)$, -2, -1, 1, 2 |
| 4. 3, -25 | 9. -12 |
| 5. $x(x-2)^2$, 0, 2 | 10. $-\infty, -\infty$ |

How to grade this. End behavior is two claims, not one: the degree's parity says whether the ends agree, and the leading coefficient's sign says which way. A student who names only one of the two has not justified the answer even when the arrows are right.

1. End behavior of $f(x) = 2x^3 - 5x + 1$.

Solution: Degree three (odd) with leading coefficient 2 (positive). Odd degree means the two ends disagree; a positive leading coefficient sends the right end up. The lower-degree terms are swamped for large $|x|$, so

$$f(x) \rightarrow +\infty \text{ as } x \rightarrow +\infty$$

$$f(x) \rightarrow -\infty \text{ as } x \rightarrow -\infty$$

2. End behavior of $g(x) = -x^4 + 3x^2$.

Solution: Degree four (even) with leading coefficient -1 (negative). Even degree means the ends agree, and the negative sign sends both of them down, so the answer is the same in both directions:

$$g(x) \rightarrow -\infty \text{ as } x \rightarrow +\infty$$

$$g(x) \rightarrow -\infty \text{ as } x \rightarrow -\infty$$

Yes, the ends match — that is exactly what even degree means.

3. Zeros and behaviour of $f(x) = (x+3)(x-1)^2$.

Solution: Set each factor to zero. $x+3=0$ gives $x=-3$ with multiplicity one, so the graph crosses there. $(x-1)^2=0$ gives $x=1$ with multiplicity two, so the graph touches the axis and turns back:

$$x = -3 \text{ (crosses), } x = 1 \text{ (touches)}$$

4. For $f(x) = (x+3)(x-1)^2$: find $f(0)$ and $f(-4)$.

Solution: The y -intercept: $f(0) = (0+3)(0-1)^2 = 3 \cdot 1$, so $f(0) = 3$.

To the left of the smallest zero, test $x = -4$: $f(-4) = (-4+3)(-4-1)^2 = (-1)(25)$, so $f(-4) = -25$. The function is negative on $(-\infty, -3)$, which agrees with the odd-degree, positive-leading-coefficient end behaviour.

5. Factor $p(x) = x^3 - 4x^2 + 4x$ and list its zeros.

Solution: Take out the common factor first, then recognise the perfect-square trinomial:

$$x^3 - 4x^2 + 4x = x(x^2 - 4x + 4) = \boxed{x(x - 2)^2}$$

The zeros are therefore $\boxed{x = 0, x = 2}$. The zero at $x = 2$ has multiplicity two, so the graph touches the axis there and turns back; at $x = 0$ it crosses.

6. End behavior of $h(x) = -3x^5 + 2x^2 - 7$.

Solution: Degree five (odd), leading coefficient -3 (negative). Odd degree makes the ends disagree, and the negative coefficient flips the standard odd picture, so the right end falls and the left end rises:

$$\boxed{h(x) \rightarrow -\infty \text{ as } x \rightarrow +\infty} \quad \boxed{h(x) \rightarrow +\infty \text{ as } x \rightarrow -\infty}$$

7. For $f(x) = (x + 2)(x - 1)(x - 4)$, evaluate $f(0)$ and $f(2)$.

Solution: The zeros -2 , 1 and 4 cut the axis into four intervals, and the sign is constant on each. Test inside two of them.

$$f(0) = (2)(-1)(-4) \Rightarrow \boxed{f(0) = 8}$$

so $f > 0$ on $(-2, 1)$.

$$f(2) = (4)(1)(-2) \Rightarrow \boxed{f(2) = -8}$$

so $f < 0$ on $(1, 4)$. All three zeros have multiplicity one, so the sign flips at each of them, which is consistent with these two test values.

8. Factor $q(x) = x^4 - 5x^2 + 4$ and list all four zeros.

Solution: Treat it as a quadratic in x^2 : it factors as $(x^2 - 1)(x^2 - 4)$, and each of those is a difference of squares:

$$x^4 - 5x^2 + 4 = \boxed{(x - 1)(x + 1)(x - 2)(x + 2)}$$

In increasing order the zeros are $\boxed{x = -2, -1, 1, 2}$, each of multiplicity one, so the graph crosses at all four.

9. Sketch $f(x) = (x + 2)^2(x - 3)$. (*The drawn curve is manual review — check it against the listed features.*)

Solution: Expanding mentally, the leading term is x^3 : degree three, positive leading coefficient, so the curve falls on the left and rises on the right. The zero $x = -2$ comes from a squared factor (multiplicity two): the graph touches the axis and turns back without changing sign. The zero $x = 3$ is simple, so the graph crosses there. The y -intercept is $f(0) = (2)^2(-3)$, that is $\boxed{f(0) = -12}$.

Putting it together: coming up from $-\infty$ on the far left, the curve rises to touch the axis at $x = -2$, turns back down, passes through $(0, -12)$, reaches a minimum, then crosses upward at $x = 3$ and rises. The function is negative on $(-\infty, 3)$ apart from the single touch point, and positive on $(3, \infty)$.

10. The claim “ $f(x) = -2x^4 + x^3$ has even degree, so both ends rise.” (*Explanation is manual review.*)

Model answer: The parity of the degree only says whether the two ends *agree*; it never says which direction they go. That is the leading coefficient’s job. Here the degree is even, so the ends do agree — but the leading coefficient is -2 , which is negative, so both ends fall rather than rise:

$$\boxed{f(x) \rightarrow -\infty \text{ as } x \rightarrow +\infty} \qquad \boxed{f(x) \rightarrow -\infty \text{ as } x \rightarrow -\infty}$$

Full credit needs both halves named: even degree (ends agree) and negative leading coefficient (both go down).